PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 08 – Polymorphism in Java

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1. Task08.md

Aim

To demonstrate runtime polymorphism in Java using the Campus Smart-Parking System.

Polymorphism

Polymorphism means one method can perform different actions depending on the object.

Types of Polymorphism

1. Compile-time polymorphism – Method overloading.
2. Runtime polymorphism – Method overriding.

Application

In the parking system, different vehicle types have different parking fees. A common Vehicle reference can refer to Car, TwoWheeler and EV objects. The overridden method is selected at runtime.

Advantages

- Code reusability
- Flexibility
- Easy maintenance
- Supports dynamic method dispatch

Conclusion

Runtime polymorphism allows the parking system to calculate different fees using a common parent-class reference.

2. FeeDemo.java

```
class Vehicle {
    double calculateFee() {
        return 30;
    }
}
class Car extends Vehicle {
    @Override
    double calculateFee() {
        return 50;
    }
}
class TwoWheeler extends Vehicle {
    @Override
    double calculateFee() {
        return 20;
    }
}
class EV extends Vehicle {
    @Override
    double calculateFee() {
        return 40;
    }
}
public class FeeDemo {
    public static void main(String[] args) {
        Vehicle[] vehicles = {
            new Vehicle(),
            new Car(),
```

```
            new TwoWheeler(),
            new EV()
        };
        String[] types = {
            "General Vehicle",
            "Car",
            "Two-Wheeler",
            "Electric Vehicle"
        };
        System.out.println("Campus Parking Fee Details:");
        for (int i = 0; i < vehicles.length; i++) {
            System.out.println(
                types[i] + " Fee: Rs. " +
                vehicles[i].calculateFee()
            );
        }
    }
}
```

3. Output:

```
Campus Parking Fee Details:

General Vehicle Fee: Rs. 30.0

Car Fee: Rs. 50.0

Two-Wheeler Fee: Rs. 20.0
```